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Small particle reagent (SPR) method for detection of latent fingermarks: A review

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Small particle reagent;
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Suspension material

Abstract Small particle reagent technique is a means to develop latent fingermarks on wet, non-porous surfaces including glass, plastic, metals and adhesive sides of tape. The method is based on the adherence of fine particles of treating solution to the oily or fatty components of latent fingermark residues. The effectiveness of technique can be improved by adding fluorescent dye in treating solution. Fluorescent small particle reagent composition not only detects faint and weak prints but it also develops latent fingermarks on multi-colored surfaces. Small particle reagent technique is convenient, efficient and cost-effective method to develop latent fingermarks on wide range of substrates of forensic importance. Standardized testing of SPR formulation is also suggested.
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1. Introduction

Fingerprints are one of the most valuable evidence due to their uniqueness. They are found on objects present at crime scene and are used to identify the suspect or criminal or link them to crime scene and weapon or object. Fingermarks are formed by sweat released from pores present on friction ridge skin of hands. Finger ridges contain large number of sweat pores. When the finger touches any surface, the sweat from these pores gets deposited in form of contours which are the mirror image of the ridge patterns. Since sweat is colorless in nature, its deposition on surface also produces colorless impressions and these impressions are called latent fingerprints.[1]

Latent fingerprint residues consist of secretions of the eccrine (sweat), sebaceous and apocrine glands present on the palm, head and nose. Sweat contains water (>98%), minerals (0.5%) and organic compounds (0.5%). Eccrine sweat consists of proteins, urea, amino acids, uric acid, lactic acid, sugars, creatinine, choline while sebaceous sweat consists of glycerides, fatty acids, wax esters, squalene and sterol esters. Different kinds of optical, physical and/or chemical methods are routinely used to visualize latent fingerprints. These methods can be used alone or in combination with others to enhance the visibility of developed prints. Selection of method depends on nature (porous, semi-porous, non-porous), color and condition (wet or dry) of surface.[1]

Small particle reagent (SPR) technique is an effective method to develop latent fingermarks on moist, non porous surfaces. The method is based on the adherence of fine particles of treating solution to the oily or fatty components of latent fingermark residues. It interacts with the water insoluble sebaceous components of latent fingermark residues. Conventionally, SPR consists of a suspension of fine molybdenum disulfide particles in an aqueous medium containing detergent solution (as surfactant). These particles adhere to fatty components of latent fingermark residues and form a gray deposit. This method is also considered as wet powdering method. This method can be used to develop latent fingermarks on wide range of surfaces such as plastic, wood, vinyl, glass, metal and cardboard which are immersed in water for long period of time.[1,2]

2. The reagent

Surfactant and suspension material are the main components of conventional SPR composition. Surfactant is a key ingredient of the SPR compositions. It is a surface active agent. It reduces the surface tension of water and enhances the moisture on surface. It also uniformly distributes the suspension material of moist surface.[2] Surfactant is a synthetic detergent which contains certain organic compounds which can be harmful for users or environment. The quality of developed prints varies with the concentration and solubility of surfactant in SPR composition. Therefore, its concentration should be optimized because high concentration can cause development of weak prints.[3] Suspension material adheres to fatty components of residue and provides color to them. Black charcoal powder or molybdenum disulfide particles are used for developing latent fingermarks on light surfaces as it forms gray deposits while titanium dioxide or zinc carbonate particles are used for developing latent fingermarks on dark surfaces as it forms white deposits.[1]

3. Methodology

Suspension material and surfactant are important ingredients of aqueous SPR composition. The effectiveness of technique, contrast and quality of developed prints depends on composition of SPR. In order to prepare SPR composition, dissolve the suspension material in distilled water and add 3 drops of surfactant to it and shake the solution well to ensure that suspension material has gone into solution. There should be no clumps of suspension material in treating solution.[1]

The working solution can be applied to suspected surface either by spraying or dipping method depending upon size and shape of object on which fingerprint has to be developed. Spraying method is best for treating large objects while dipping method is best for treating small objects. The gentle washing of treated surface with distilled water is required in order to remove excess of powder particles from them.[1] Different methods (spraying, immersion, dipping, pasting, pouring) of applying SPR-White (SPR-W) was evaluated and it was observed that submersion was most effective method for developing latent fingermarks on electrical and duct tape.[4] The wiping of SPR treated non-porous substrates with paper towel was recommended in order to improve the effectiveness of SPR process in significant manner.[5] Pre-treatment of surface with volatile bases such as ethanolamine or triethylamine was recommended before development of latent fingermarks on acidified non-porous substrates with SPR or powder dusting procedures.[6]

4. Evaluation of SPR method

The use of tergitol-7 detergent in combination with choline chloride in molybdenum disulfide based SPR composition was suggested.[3] It was also observed that choline chloride was not an essential component of SPR composition and use of suspension of molybdenum disulfide dispersed in Manoxol OT based SPR composition was suggested.[7] Another study recommends the use of tergitol-7 in zinc carbonate based modified SPR composition.[8] Generally, surfactant is a synthetic detergent and contains certain organic compounds which can be harmful for users or environment.[3] The use of Saponin (natural detergent), in place of synthetic detergent, in two SPR compositions based on charcoal powder and basic zinc carbonate was suggested to develop latent fingermarks on a wide range of surfaces including compact discs due to its durable shelf life and high quality of developed prints.[9]

Use of xerox powder, lead oxide, cobalt oxide and graphite in place of molybdenum disulfide particles was also suggested.[3] Iron oxide based SPR composition was found to be more effective than conventional molybdenum disulfide based SPR composition and was recommended for developing latent fingermarks on surfaces of smooth surface[10] but another study contradicted the iron oxide based SPR due to its low sensitivity as compared to molybdenum disulfide based composition.[11] A titanium dioxide based SPR in tergitol was also recommended for developing latent fingermarks on wet plastic bottle.[12] Some authors used titanium dioxide based SPR composition to develop latent fingermarks on glass, plastic and metal

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surfaces.[13] The rutile form of titanium dioxide was used in SPR composition to develop latent fingermarks on dark, wet surfaces and on sticky side of the black electric tape.[14] Titanium dioxide based SPR composition containing Kodak photo-flo 200 was also used to develop latent fingermarks on both adhesive and non-adhesive sides of tape.[4] Black and silver fingerprint powder based SPR compositions in terizide (dish washing liquid) was also reported.[15] The use of SPR technique and some lysochromes (sudan black and oil red O) was recommended to develop fresh and aged (up to 15 days) latent fingermarks on wet glass and plastic surfaces.[16] Silver-gray SPR composition was recommended to develop latent fingermarks on simulated vehicle-borne improvised explosive devices.[45]

In order to resolve the problem of contrast on dark surfaces with conventional SPR composition, zinc carbonate based composition was recommended for developing prints on dark and wet surfaces. It was observed that adherence of powder on latent fingermarks and quality of developed marks depend on the particle size of zinc carbonate powder. Particles of 2 μm average size are recommended because they adhere better than those of 6 μm average size particles.[8] Particle size of the suspension material in SPR composition affects the quality of the developed prints. It was observed that smaller particle size gave unsatisfactory results while larger particle size gave satisfactory results.[17] Tergitol-7 along with ROCOL™ as powder was also used to study the effect of structure of molybdenum disulfide on the quality of developed prints.[18] Due to significant effectiveness and better quality of developed prints, zinc oxide based SPR composition was suggested over dry powder composition to develop latent fingermarks (fresh and aged) on a variety of non-porous substrates. It was observed that lithium doped zinc oxide gave more fluorescence than pure zinc oxide.[19] Effect of topography of substrate surface was studied using iron oxide based powder suspension to develop latent fingerprints on smooth plastic surface and observed that topographical feature, shape and roughness of surface effects the processing of latent fingermarks and quality of developed prints.[20]

Conventional SPR composition is less effective in developing latent fingermarks on multi-colored surfaces due to poor contrast. Its effectiveness further reduces with rise in immersion time of object/s in water. To resolve these problems, fluorescent SPR compositions containing basic yellow 40 and rhodamine 6G were recommended. Use of fluorescent dyes along with molybdenum disulfide in SPR compositions to develop latent fingermarks on moist, multi-colored surfaces was suggested. Zinc carbonate based SPR composition was also used because zinc carbonate causes negligible effect on resulting fluorescence.[21] SPR compositions containing fluorescent dyes have been studied and effectiveness of the developed prints with respect to immersion time and shelf life of the reagents have also been reported. A fluorescent cyano blue based SPR was recommended for developing latent fingerprints on multi-colored, wet, non porous surfaces.[22] A fluorescent SPR composition based on basic zinc carbonate and crystal violet in commercial liquid detergent was also proposed to develop latent fingerprints on wet, non porous surfaces.[23] The use of basic fuchsin based SPR composition for developing aged (45 day old) latent fingerprints on moist non-porous surfaces was suggested. The use of basic fuchsin was advocated over crystal violet based composition due to high quality of developed prints with former one.[24] A fluorescent SPR composition containing basic zinc carbonate, eosin Y dye and commercial liquid detergent was also suggested to develop aged latent fingermarks (upto 36 h) on wide spectrum of wet, multi-colored smooth surfaces.[25] In another study, a fluorescent SPR composition containing basic zinc carbonate, rhodamine B dye and commercial liquid detergent was suggested over conventional molybdenum disulfide based SPR composition to develop latent fingermarks on a wide spectrum of wet, non-porous surfaces.[26] In another study, a fluorescent SPR composition containing basic zinc carbonate, eosin B dye and commercial liquid detergent was recommended to develop latent fingermarks on aluminum foil, lamination sheets and compact discs which had been immersed in water and wet soil for varying period of time.[27] In a similar study, SPR composition containing basic zinc carbonate, crystal violet dye and commercial liquid detergent was used for developing fresh and aged latent fingermarks on different kinds of moist, smooth surfaces. It was observed that quality of water (fresh or stagnant) affects the quality of developed prints.[28] In an analogous study, a fluorescent SPR composition containing basic zinc carbonate along with titanium dioxide and zinc stearate, brilliant blue R dye and commercial liquid detergent was used for developing fresh and aged (upto 10 days) latent fingermarks on moist surfaces of non-porous nature.[29]

Different kinds of processing methods such as black powdering, iodine fuming, cyanoacrylate fuming and SPR were used to develop latent fingerprints on writing surface of compact discs and observed that after treatment with these techniques, recovery and rewriting of data on these treated compact discs vary with brands.[30] In another study, a fluorescent SPR composition containing basic zinc carbonate, eosin Y dye and commercial liquid detergent was used to develop latent fingermarks on writing surface of different kinds of rewriteable and recordable compact discs. It was observed that development process neither despoiled nor damaged the stored data and it could be retrieved from different discs. There was also no problem in saving additional data on these compact discs.[31]

SPR composition consists of basic zinc carbonate and eosin B was suggested to develop latent fingermarks on different kinds of non-porous surfaces which were exposed to high temperature (~900 °C) for 1 hour before processing. It was observed that eosin B based composition gave better quality, intense, fluorescent prints as compared to eosin Y based SPR composition.[32] In a similar study, SPR compositions containing basic zinc carbonate and eosin B and eosin Y dyes in commercial liquid detergent were suggested for enhancing bloodied marks on a variety of smooth surfaces which were exposed to a high temperature (~900 °C) for 1 hour before processing.[33] A comparative study reveals the use of iron oxide and Kodak photo-flo based SPR composition over cyanoacrylate fuming and vacuum metal deposition methods for developing latent fingermarks on wet non-porous surfaces recovered from fire scene.[34]

The use of titanium dioxide based wet powder suspension was suggested for developing bloodied marks on dark, smooth non-porous surfaces. It was observed that this composition did not interfere in subsequent presumptive tests for blood. Incorporation of this technique into standard protocol for enhancement of bloodied prints was also suggested. It was recommended that this procedure should be followed only when subsequent DNA profiling from blood is not required as it reduces the amount of DNA recoverable from developed

prints.[35] It is also observed that SPR technique does not cause any interference in subsequent STR typing from blood stains.[36]

SPR method is more effective than cyanoacrylate method for the development of latent fingerprints on water soaked firearms and it does not cause any interference in any subsequent analysis of firearms.[37] It is observed that cyanoacrylate fuming is a better method than SPR for developing latent fingermarks on dry surfaces.[12] It is observed that white SPR method is more effective than Swedish soot mixture powder and cyanoacrylate fuming method for developing aged (7 day old) latent fingermarks on wet transparent foil.[38] Due to high quality of developed prints, black magnetic powder and black powder suspension methods were suggested to develop latent fingermarks on different kinds of food items after comparing them with cyanoacrylate fuming, ninhydrin, SPR and white powder suspension.[39] The use of powder suspension over vacuum metal deposition method was recommended to develop the latent fingermarks on moist, non-porous surfaces.[40] A comparative study revealed that cyanoacrylate fuming method is more effective and efficient than SPR and powdering methods in developing aged (upto 168 h) latent fingerprints on glass and metal surfaces immersed in stagnant water. It was observed that quality of developed fingerprints depends on method of enhancement and duration of exposure to stagnant water.[41] Another comparative study suggested the use of zinc carbonate and cyano blue dye based SPR composition over other processing techniques (iodine fuming, cyanoacrylate fuming, powder dusting) for developing latent fingermarks on both emulsion and non-emulsion sides photographic film. Fresh fingermarks responded better than aged fingermarks to SPR processing. Striking difference was observed in the quality of developed prints on both sides of photographic film.[42] A comparative study suggested the use of SPR technique to study the effect of common household liquids (wine, beer, milk, soft drinks, soapy water and orange juice) on latent fingermarks on a non-porous surface.[43] In a similar comparative study, the use of SPR-W was recommended over amido black 10b and acid yellow 7 for developing fresh and aged (3 weeks) blood prints on black-handled knives.[44]

5. Conclusion

SPR technique is a convenient, cost-effective and efficient methodology to develop latent fingermarks on wide range of wet, non-porous surfaces present at crime scene. Incorporation of fluorescent dyes in SPR composition not only detects weak and faint fingermarks but it also detects fingermarks on multicolored surfaces. The present technique is simple, effective and even an amateurish hand can operate it without any requirement of specialized equipment. Standardized testing of SPR formulation should be conducted to evaluate its efficiency and reliability in developing moist latent fingermarks on different surfaces. More research is required to better understand the reaction mechanism involved and to improve the sensitivity of this for processing latent fingermarks (aged and degraded) on difficult and unusual surfaces.

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Conflict of interest

None to declare.

Ethical approval

Necessary ethical approval was obtained from the institute ethics committee.

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